

Contents

Bayesian Estimation	1
Example 1: Coin Toss	1
Example 2: Normal Distribution	2

Bayesian Estimation

EEE 554 Lecture #15

In Bayesian estimation, the parameter θ is taken to be a random variable (RV) Θ with a known PDF $f_{\Theta}(\theta)$, often called the “prior” PDF.

The Bayesian estimator for θ is the RV $\hat{\Theta}_B$ with the posterior PDF:

$$f_{\Theta}(\theta|\underline{x}) = \frac{f_{\underline{X}}(\underline{x}|\theta)f_{\Theta}(\theta)}{f_{\underline{X}}(\underline{x})}$$

Both factors in the numerator (i.e., the likelihood function and the prior PDF of θ) are assumed known; the denominator is obtained by marginalization:

$$f_{\underline{X}}(\underline{x}) = \int_{-\infty}^{\infty} f_{\underline{X},\Theta}(\underline{x},\theta)d\theta$$

where:

$$f_{\underline{X},\Theta}(\underline{x},\theta) = f_{\underline{X}}(\underline{x}|\theta)f_{\Theta}(\theta)$$

Example 1: Coin Toss

The probability of a coin toss yielding Heads (H) is p . * **Data:** $E = \{n \text{ out of } n \text{ independent tosses yield H}\}$
* **Prior:** Take $p \sim U[0, 1]$.

Using Bayes rule:

$$f_P(p|E) = \frac{P(E|p)f_P(p)}{P(E)}$$

with $P(E|p) = p^n$ and $f_P(p) = 1$ for $0 \leq p \leq 1$ (0 otherwise).

So,

$$f_P(p|E) = \frac{p^n}{P(E)} \quad \text{for } 0 \leq p \leq 1$$

For proper normalization:

$$1 = \int_0^1 f_P(p|E)dp = \frac{1}{P(E)} \int_0^1 p^n dp = \frac{1}{(n+1)P(E)}$$

and hence $P(E) = \frac{1}{n+1}$ and $f_P(p|E) = (n+1)p^n$ for $0 \leq p \leq 1$.

The Bayesian estimator thus has mean:

$$E[\hat{P}_B] = E[P|E] = \int_0^1 p f_P(p|E)dp = \int_0^1 p(n+1)p^n dp = \frac{n+1}{n+2}$$

Note that $E[\hat{P}_B] \rightarrow 1$ as $n \rightarrow \infty$, as expected. Also, it is possible to compute $P(E)$ directly if desired:

$$P(E) = \int_0^1 P(E|p)f_P(p)dp = \int_0^1 p^n \cdot 1dp = \frac{1}{n+1}$$

Example 2: Normal Distribution

Let $X_1, \dots, X_n$ be iid $\mathcal{N}[\theta, 1]$ RVs. The parameter to be estimated is $\theta = \mu$. Take the prior PDF of θ to be $\mathcal{N}[0, \rho^2]$.

Then:

$$f_{\underline{X}}(\underline{x}|\theta) = \frac{1}{(2\pi)^{n/2}} \exp \left\{ -\frac{1}{2} \sum_{k=1}^n (x_k - \theta)^2 \right\}$$

and

$$f_{\Theta}(\theta) = \frac{1}{\sqrt{2\pi\rho^2}} \exp \left\{ -\frac{1}{2} \frac{\theta^2}{\rho^2} \right\}$$

Denoting $C = \frac{1}{(2\pi)^{n/2}} \cdot \frac{1}{\sqrt{2\pi\rho^2}} \cdot \frac{1}{f_{\underline{X}}(\underline{x})}$:

$$\begin{aligned} f_{\Theta}(\theta|\underline{x}) &= C \exp \left\{ -\frac{1}{2} \sum_{k=1}^n (x_k - \theta)^2 \right\} \exp \left\{ -\frac{1}{2} \frac{\theta^2}{\rho^2} \right\} \\ &= C \exp \left\{ -\frac{1}{2} \left[\sum_{k=1}^n x_k^2 - 2\theta \sum_{k=1}^n x_k + n\theta^2 + \frac{\theta^2}{\rho^2} \right] \right\} \\ &= C \exp \left\{ -\frac{1}{2} \sum_{k=1}^n x_k^2 \right\} \exp \left\{ -\frac{1}{2} \left[\left(n + \frac{1}{\rho^2} \right) \theta^2 - 2\theta \sum_{k=1}^n x_k \right] \right\} \\ &= C \exp \left\{ -\frac{1}{2} \sum_{k=1}^n x_k^2 \right\} \exp \left\{ -\frac{1}{2} \left(n + \frac{1}{\rho^2} \right) \left[\theta^2 - 2 \frac{\sum_{k=1}^n x_k}{n + \frac{1}{\rho^2}} \theta \right] \right\} \end{aligned}$$

Completing the square in θ :

$$f_{\Theta}(\theta|\underline{x}) = C \exp \left\{ -\frac{1}{2} \sum_{k=1}^n x_k^2 \right\} \exp \left\{ \frac{1}{2} \frac{(\sum_{k=1}^n x_k)^2}{n + \frac{1}{\rho^2}} \right\} \times \exp \left\{ -\frac{1}{2} \left(n + \frac{1}{\rho^2} \right) \left[\theta - \frac{\sum_{k=1}^n x_k}{n + \frac{1}{\rho^2}} \right]^2 \right\}$$

This represents a normal PDF with mean $\frac{\sum x_k}{n+1/\rho^2}$, hence:

$$E[\hat{\Theta}_B] = \frac{1}{n + \frac{1}{\rho^2}} \sum_{k=1}^n x_k$$

Note that as $\rho^2 \rightarrow \infty$, the prior becomes uninformative and:

$$E[\hat{\Theta}_B] \rightarrow \frac{1}{n} \sum_{k=1}^n x_k$$

(i.e., the sample mean).

As $\rho^2 \rightarrow 0$, the prior dominates the data and $E[\hat{\Theta}_B] \rightarrow 0$ regardless of $\underline{x}$.

It is helpful to observe that $f_{\underline{X}}(\underline{x})$ and several other normalizing values never had to be calculated. Because $f_{\Theta}(\theta|\underline{x})$ was known to be a PDF in this example, only the exponential containing θ was needed to deduce the mean.